

LESSON 7.5

GRAPHING LINES FOR REAL-WORLD CONTEXTS



LESSON INTRODUCTION

MA.8.AR.3.4 Given a mathematical or real-world context, graph a two-variable linear equation from a written description, a table, or an equation in slope-intercept form.

LEARNING TARGETS

- I can graph a two-variable linear equation from an equation, table, or written description given a real-world context.

BENCHMARK COHERENCE

BUILDING ON

MA.7.AR.4.3 Given a mathematical or real-world context, graph proportional relationships from a table, equation, or a written description.

WORKING TOWARD

MA.912.AR.2.4 Given a table, equation, or written description of a linear function, graph that function and determine and interpret its key features.

ESSENTIAL QUESTIONS

- How can I use the information given from a written description to graph a linear relationship?
- How can I use a linear equation or table of values to graph a linear relationship?

LESSON NARRATIVE

Students make connections between graphing two-variable linear equations and real-world situations. Students graph lines given linear relationships represented as equations, tables, and written descriptions in a real-world context.

COMPONENT SUMMARY

7.5.1 WARM-UP (~5 MIN.)

Determine which graph does not belong with the others and justify their reasoning

7.5.3 BRING IT TOGETHER (5 MIN.)

Determine what information is needed to graph a linear relationship from a real-world context

7.5.5 YOUR TURN! (10 MIN.)

Additional practice graphing lines from written descriptions and tables of values of real-world linear relationships

7.5.2 EXPLORATION (10-15 MIN.)

Explore how to graph lines from written descriptions of real-world linear relationships

7.5.4 GUIDED PRACTICE (10-15 MIN.)

Practice graphing lines from written descriptions and tables of values of real-world linear relationships

7.5.6 WRAP-UP (~5 MIN.)

Demonstrate understanding of graphing linear relationships from a real-world context

RESOURCES

GLOSSARY

Key Terms: N/A

Additional Terms:

- Slope-intercept form
- Linear relationship
- Two-variable linear equation
- Slope
- y-intercept

MATERIALS

- Straightedge
- (Optional) Desmos or graphing calculators

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may struggle deciding how to label the axes when graphing real-world relationships.
- Students may not know what information they need from the written description to graph the line.

THINGS TO CONSIDER

- Graphs are limited to Quadrant 1 for this lesson. The teacher can facilitate discussion on why the x- and y- values are both positive based on the contexts addressed. This will be developed further in the next lesson.
- Emphasize the importance of always including units. Thinking in terms of the units throughout the problem-solving process will help students understand the context of the relationship.

LITERACY AND LANGUAGE ROUTINES

Which One Doesn't Belong?

The 7.5.1 Warm-Up provides students with the opportunity to choose which graph they believe does not belong with the others. All of the options can be selected, so there is no wrong answer. This encourages all students to participate. The focus is having students justify why the graph they selected does not belong.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.7.1 Real-World Contexts

Students graph linear representations of real-world contexts. They determine what information they need from written descriptions to graph the line. For example, in the 7.5.2 Exploration, students must determine the two points given in a written description and then graph the points and connect them to represent the linear relationship.

DIFFERENTIATE INSTRUCTION

7.5.2 Exploration, 7.5.4 Guided Practice, and 7.5.5 Your Turn! require students to determine what information they need in a written description to graph the linear relationship. Have students underline, circle, or label the key information. Consider adding scaffolding questions to the 7.5.5 Your Turn! activity, similar to what is included in the 7.5.4 Guided Practice, to help students who are struggling determine the important information. Also, consider having students check their graphs using Desmos or graphing calculators.

7.5.1 WARM-UP: WHICH ONE DOESN'T BELONG?

TEACHER MOVES

Encourage students to use mathematical vocabulary when justifying their choice. Involve several students, ideally students who rarely participate. Participating has the potential to improve students' efficacy and agency.

ENRICHMENT

- Who used slope as the criteria to determine which graph doesn't belong? (Call on a student to share.)
- Who used the y-intercept as the criteria to determine which graph doesn't belong? (Call on a student to share.)
- What other criteria can you use to mathematically justify which one does not belong with the others?

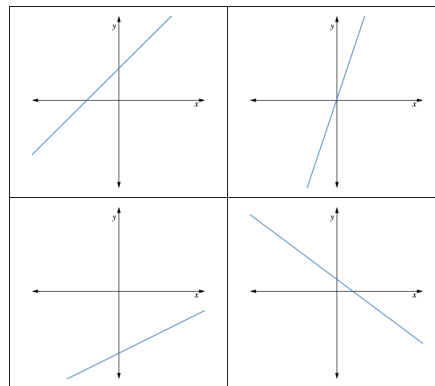
QUESTIONING

Challenge students who finish early to think of a reason (and even multiple reasons) that would justify why each of the options do not belong. This will keep all students engaged while students who work at a slower pace determine one response.



7.5.1 Warm-Up: Which One Doesn't Belong?

The graphs of four linear relationships are shown.



1. Circle the graph that doesn't belong. Explain your reasoning.

Answers will vary. An example would be the bottom right does not belong because it is the only one with a negative slope.



7.5.2 Exploration: Graphing Linear Relationships for Real-World Contexts

- When plugged into a USB charger, cell phones charge at a constant rate until they get closer to being fully charged.

When Michaela plugged in her phone, the battery was at 16%. After charging for 15 minutes, the battery level was at 41%.

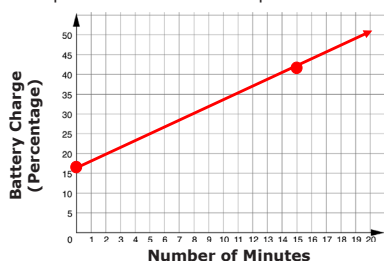
This linear relationship can be represented by a graph where x represents the number of minutes the phone was plugged in, and y represents the percentage of the battery charged.

Harry said there is enough information from this context to find two points on the graph of this relationship.

- Work with your partner to determine the coordinates of the two points.

Which point represents the phone's charge when Michaela plugged it in?	Which point represents the phone's charge 15 minutes after Michaela plugged it in?
$(0, 16)$	$(15, 41)$

- Graph the linear relationship.



- What information was used to graph the relationship?
The two points from the context were used to graph the relationship.
- Explain why the slope was not needed to graph this relationship.
The slope between any two points on a line is the same, only two points were needed.

- The body length, L , of a milk snake in inches can be represented by the equation $L = 6.5 + 2m$, where m is the age of the snake in months for the first two years of its life.



- Graph the relationship between L and m .



- What information was used to graph the relationship?
The slope and y-intercept.
- Discuss with your partner how to determine from the equation which value represents the slope and which value represents the y-intercept. Write a brief summary of your discussion.
The slope is being multiplied by the independent variable. The y-intercept is the constant value.

7.5.2 EXPLORATION: GRAPHING LINEAR RELATIONSHIPS FOR REAL-WORLD CONTEXTS

TEACHER MOVES

Consider encouraging students to highlight, underline, circle, or label the key information as a form of annotation while they read. Also, use a Think Aloud to help students understand how you determined the key information and used it to graph the line.

ENRICHMENT

How did you determine the points on question 1? How do you know which quantity is x and which is y ? Besides being a point on the line, what does $(0,16)$ represent?

QUESTIONING

Consider asking students who finish early to estimate how many minutes it would take to fully charge the battery based on the information provided.

TEACHER MOVES

Compare the information used to create each graph. Discuss what specific information provided would lead to using a specific method for graphing. These discussions will improve student understanding and provide auditory entry points.

7.5.3 BRING IT TOGETHER: GRAPHING RELATIONSHIPS FROM REAL-WORLD CONTEXTS

DIFFERENTIATE INSTRUCTION

Consider having students record this information on a reference card to use in subsequent components.



7.5.3 Bring It Together: Graphing Relationships from Real-World Contexts

For each context, different information was used to graph the linear relationships.

To graph a linear relationship given a real-world context, the information that is necessary to know are either

- the slope only
- the y-intercept only
- the slope and y-intercept

or

- two ordered pairs.
- a table of values with five ordered pairs.

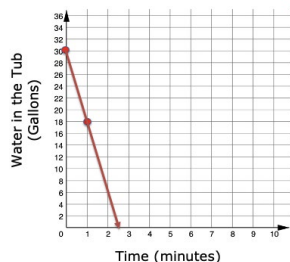


7.5.4 Guided Practice: Graphing Linear Relationships for Real-World Contexts

1. A bathtub drains at a constant rate. There are 30 gallons of water in the tub when Camille opens the drain. The bathtub drains at 12 gallons per minute.
 - a. Which information about the relationship can be determined from the situation?
 - the slope and y-intercept of the graph
 - at least two ordered pairs on the graph
 - b. What are the slope and y-intercept, or at least two of the ordered pairs?

slope = $-\frac{12 \text{ gallons}}{1 \text{ minute}}$ and y-intercept = 30 gallons

- c. Graph the linear relationship between time, in minutes, x , and the number of gallons in the bathtub, y .



7.5.4 GUIDED PRACTICE: GRAPHING LINEAR RELATIONSHIPS FOR REAL-WORLD CONTEXTS

ENRICHMENT

How did you know that context provided information about the slope and y-intercept for question 1?

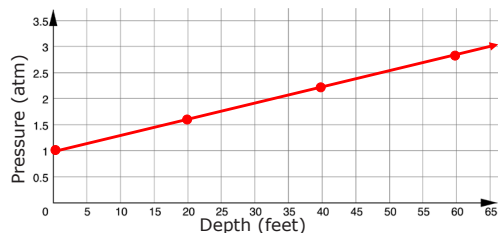
QUESTIONING

Ask students to interpret the meaning of the slope, y-intercept, and a point based on the context.

2. Scuba divers wear protective equipment when diving because the pressure placed on the body increases as they dive, due to the weight of the water. At sea level, a diver experiences 1 atmosphere of pressure (atm). The table shows the pressure on a scuba diver at various diving depths.

Depth (feet)	0	20	40	60
Pressure (atm)	1	1.6	2.2	2.8

- a. Graph the linear relationship between depth in feet, x , and pressure, in number of atmospheres of pressure (atm), exerted on the diver, y .



- b. What is the slope of the line?

$$\text{Slope} = \frac{1.6-1}{20-0} = \frac{0.6}{20} = 0.03$$

ENRICHMENT

- On question 2, did you use all the points provided? Why or why not?
- Which points did you choose to graph and why?

7.5.5 YOUR TURN!

TEACHER MOVES

Consider having students discuss their work with a partner after completing the questions independently.

ENRICHMENT

- How did you determine what information to use?
- Did you need to use all the information provided?
- How did you determine the y-intercept and slope?
- What does each data point represent in context?

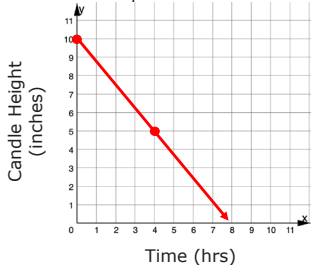


7.5.5 Your Turn!

1. A candle is 10 inches long. Ximena lights the candle and records the length of the candle, y , in inches (in.) over x hours (hrs). The table shows Ximena's recordings.

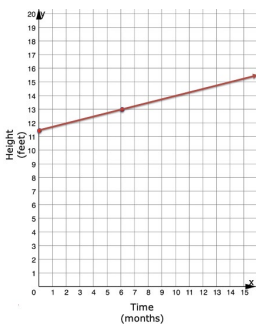
Time (hrs)	0	1	3	4
Candle height (in.)	10	8.75	6.25	5

Graph the relationship between candle height and time.



2. Gabriel planted a Leyland Cypress tree in his front yard. The tree was 11.5 feet tall when he planted it. It reached a height of 13 feet after 6 months.

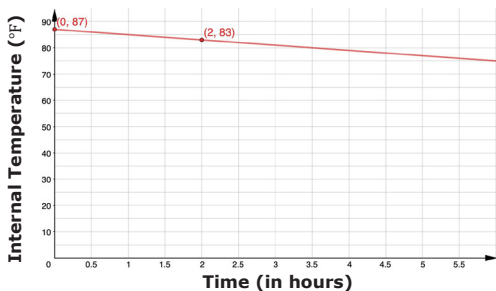
Graph the linear relationship between the height of the tree in feet, y , and the number of months since it was planted in Gabriel's yard, x .



**7.5.6 Wrap-Up: Ticket Out the Door**

The Johnstons turned their air conditioner off while they were out of town. When they returned home, the internal temperature of the house was 87°F . They turned the air conditioner back on and after 2 hours, the temperature had dropped to 83°F . The internal temperature of the house decreased at a constant rate until reaching 75°F .

1. Graph the linear relationship.

**7.5.6 WRAP-UP: TICKET OUT THE DOOR****DIFFERENTIATE INSTRUCTION**

Consider allowing students to use an audio recorder to respond to question 1.